

Symmetries and Degeneracies

This chapter applies the representation theory developed above to quantum mechanics. Its central question is very concrete:

If a Hamiltonian is invariant under a group of transformations, what can one know about its eigenvalues and eigenvectors before solving the full eigenvalue problem?

The answer separates two pieces of information. The dimensions of the irreducible representations determine the degeneracies forced by symmetry, whereas their multiplicities determine the smaller matrices that still have to be diagonalised. We first derive this statement in complete generality and then work it out for rotations, spin systems, open chains, and rings.

23.1 Quantum transformations and Wigner's theorem

Let V be a complex Hilbert space. A nonzero vector $\psi \in V$ describes a pure state, but its phase has no physical significance: ψ and $e^{i\theta}\psi$ represent the same ray. For normalised states, the transition probability is

$$P(\psi_1, \psi_2) = |\langle \psi_1, \psi_2 \rangle|^2.$$

Thus a transformation of pure states must be a bijection of rays that preserves this quantity.

Theorem 23.1: Wigner's theorem

Let $\mathcal{D}$ be a bijective transformation of the rays of V such that

$$|\langle \mathcal{D}\psi_1, \mathcal{D}\psi_2 \rangle|^2 = |\langle \psi_1, \psi_2 \rangle|^2 \quad (\psi_1, \psi_2 \in V).$$

After choosing one phase on each transformed ray, $\mathcal{D}$ can be represented either by a unitary operator or by an antiunitary operator.

Here an antiunitary operator is conjugate-linear:

$$\mathcal{D}(\alpha_1\psi_1 + \alpha_2\psi_2) = \alpha_1^*\mathcal{D}\psi_1 + \alpha_2^*\mathcal{D}\psi_2, \quad \langle \mathcal{D}\psi_1, \mathcal{D}\psi_2 \rangle = \langle \psi_1, \psi_2 \rangle^*.$$

In particular, $\mathcal{D}^\dagger \mathcal{D} = I$. A useful mnemonic for conjugate-linearity is

$$c\mathcal{D} = \mathcal{D}c^* \quad (c \in \mathbb{C}),$$

where the equality means equality of the corresponding maps.

Detailed proof idea. Choose an orthonormal basis (e_j) . Preservation of transition probabilities sends the mutually orthogonal rays $[e_j]$ to mutually orthogonal rays. Choose unit vectors f_j on those image rays. A ray with a representative

$$\psi = \sum_j c_j e_j$$

has transition probability $|c_j|^2$ to $[e_j]$, so its image has a representative

$$\psi' = \sum_j c'_j f_j, \quad |c'_j| = |c_j|.$$

Use the irrelevant overall phase to arrange $c'_1 = c_1$ whenever $c_1 \neq 0$. Comparing transition probabilities with the rays of $(e_1 + e_j)/\sqrt{2}$ determines the real parts of c'_j , while comparison with $(e_1 + ie_j)/\sqrt{2}$ determines the imaginary parts. The consistency of these comparisons leaves exactly two possibilities, the same one for every j :

$$c'_j = c_j \quad \text{for all } j, \quad \text{or} \quad c'_j = c_j^* \quad \text{for all } j.$$

The map obtained by $e_j \mapsto f_j$ is therefore respectively linear or conjugate-linear, and it preserves the inner product in the unitary or antiunitary sense. This is the essential argument; the zero coefficients are handled by continuity or by omitting them from the comparisons. ■

For a transformation group G , the operators assigned to group elements need not multiply exactly after phases have been chosen. In general,

$$\mathcal{D}(a)\mathcal{D}(b) = \omega(a, b)\mathcal{D}(ab), \quad |\omega(a, b)| = 1.$$

This is a *projective* unitary or antiunitary representation. Passing to a suitable covering group often converts a projective representation into an ordinary representation; spin representations of $SO(3)$ and ordinary representations of $SU(2)$ are the standard example.

23.1.1 Two different uses of a unitary operator

It is important not to confuse a change of description with a physical transformation of the system.

1. In a passive change of quantum description, both states and observables are transformed:

$$|\psi\rangle \mapsto \mathcal{D}|\psi\rangle, \quad A \mapsto \mathcal{D}A\mathcal{D}^\dagger.$$

Nothing measurable changes, because

$$\begin{aligned} \langle \mathcal{D}\psi | \mathcal{D}A\mathcal{D}^\dagger | \mathcal{D}\psi \rangle &= \langle \psi | \mathcal{D}^\dagger \mathcal{D}A\mathcal{D}^\dagger \mathcal{D} | \psi \rangle \\ &= \langle \psi | A | \psi \rangle. \end{aligned}$$

This viewpoint is especially useful for a time-dependent change of frame.

2. In an active transformation, the physical system is changed while the measuring apparatus is held fixed. One may transform the state alone,

$$|\psi\rangle \mapsto \mathcal{D}(a)|\psi\rangle, \quad A \text{ fixed,}$$

or, equivalently, hold the state fixed and transform the observable.

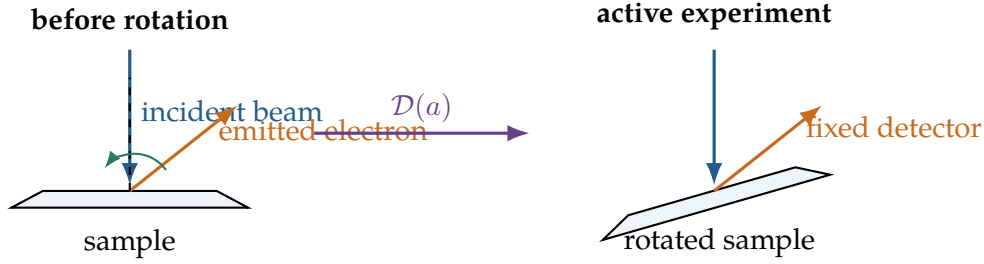


图 23.1: In photoemission, rotating the sample while leaving the incoming beam and detector fixed is a physical transformation, not merely a change of coordinates.

23.2 Hamiltonian symmetry and the first examples

Let $H : V \rightarrow V$ be a linear Hamiltonian and let

$$D : G \longrightarrow U(V)$$

be a concrete unitary representation of a group G on the physical Hilbert space. The Hamiltonian has the symmetry G when

$$D(a)HD(a)^{-1} = H \quad (a \in G),$$

or, equivalently,

$$[H, D(a)] = 0 \quad (a \in G).$$

Specifying the physical representation is essential. The trivial representation $D(a) = I$ commutes with every Hamiltonian, but it contains no information about the system.

If D is irreducible on a finite-dimensional complex space, the scalar commutant result proved in the preceding chapter immediately gives

$$H = EI.$$

Every state then has the same energy. Physical representations are more often reducible, and the purpose of this chapter is to understand the resulting block structure.

Example 23.1: Rotational symmetry of a central potential

For a particle in three dimensions with

$$H = \frac{\mathbf{p}^2}{2m} + V(r), \quad r = \|\mathbf{x}\|,$$

the rotations are represented by

$$D(\hat{\mathbf{n}}, \phi) = \exp\left(-\frac{i}{\hbar}\phi \hat{\mathbf{n}} \cdot \mathbf{J}\right).$$

Both $\mathbf{p}^2$ and r are rotational scalars, so $[H, D(\hat{\mathbf{n}}, \phi)] = 0$ for every axis and angle.

For the hydrogen atom, the familiar notation

$$E_{n\ell m} = E_n$$

contains two logically different statements. Rotational $SO(3)$ symmetry forces the $2\ell + 1$ values $m = -\ell, \dots, \ell$ at fixed (n, ℓ) to have the same energy. Equality of the energies for different values of ℓ is not forced by $SO(3)$; it is an “accidental” degeneracy from the viewpoint of that group and is explained by the larger dynamical symmetry generated by angular momentum and the Runge–Lenz vector.

Example 23.2: Parity does not force degeneracy

For the one-dimensional harmonic oscillator,

$$H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 q^2,$$

define parity by

$$(\Pi\psi)(x) = \psi(-x).$$

Then $\Pi^{-1} = \Pi^\dagger = \Pi$, $\Pi^2 = I$, and $[H, \Pi] = 0$. Hence

$$D(e) = I, \quad D(s) = \Pi$$

is an infinite-dimensional unitary representation of $\mathbb{Z}_2 = \{e, s\}$. Nevertheless, the oscillator energies

$$E_n = \hbar\omega \left(n + \frac{1}{2} \right)$$

are nondegenerate. The reason is that the irreducible complex representations of the Abelian group $\mathbb{Z}_2$ are one-dimensional. The energy eigenstates alternate between even and odd parity, but symmetry does not require two states to share an energy.

Antiunitary symmetries

Time reversal is normally antiunitary. Its invariance condition is still $\Theta H \Theta^{-1} = H$, interpreted as an equality of maps, but the commutant calculation below uses complex-linear intertwiners and therefore assumes a unitary representation. Antiunitary symmetry can lead to extra phenomena, notably Kramers degeneracy when $\Theta^2 = -I$.

23.3 The symmetry-adapted decomposition

Assume from now on that V is finite-dimensional, D is unitary, and H is linear with $[H, D(a)] = 0$ for every $a \in G$. Unitarity makes D completely reducible, so

$$D \cong \bigoplus_{\alpha=1}^k \bigoplus_{i=1}^{n_\alpha} D_i^{(\alpha)} = \bigoplus_{\alpha=1}^k n_\alpha D^{(\alpha)}.$$

Correspondingly,

$$V = \bigoplus_{\alpha=1}^k \bigoplus_{i=1}^{n_\alpha} V_i^{(\alpha)}.$$

The label α distinguishes inequivalent irreducible types. For a fixed α , the label $i = 1, \dots, n_\alpha$ distinguishes equivalent copies. Write

$$d_\alpha = \dim V_i^{(\alpha)};$$

this dimension is independent of i .

23.3.1 Constructing a common basis in the equivalent copies

Choose a basis

$$\{q_{im}^{(\alpha)} : m = 1, \dots, d_\alpha\}$$

of each $V_i^{(\alpha)}$. The union of these bases spans V . This is the representation-theoretic part of the problem and does not require knowing H . For a Lie group one often obtains these vectors by solving the simultaneous structure imposed by the Hermitian generators in

$$D(a) = \exp\left(-\frac{i}{\hbar} \eta_a^r J_r\right).$$

Equivalent copies initially need not have equal matrices. If $\tilde{D}_i^{(\alpha)}$ and $\tilde{D}_j^{(\alpha)}$ are equivalent, there is an invertible intertwiner $S_{ij}^{(\alpha)}$ such that

$$\tilde{D}_i^{(\alpha)}(a) = (S_{ij}^{(\alpha)})^{-1} \tilde{D}_j^{(\alpha)}(a) S_{ij}^{(\alpha)}.$$

Fix one reference copy j . Changing the basis in copy i by the intertwiner $S_{ij}^{(\alpha)}$ makes the matrices equal:

$$D_i^{(\alpha)}(a) = D_j^{(\alpha)}(a) = D^{(\alpha)}(a) \quad \text{for every } i, j \text{ and } a.$$

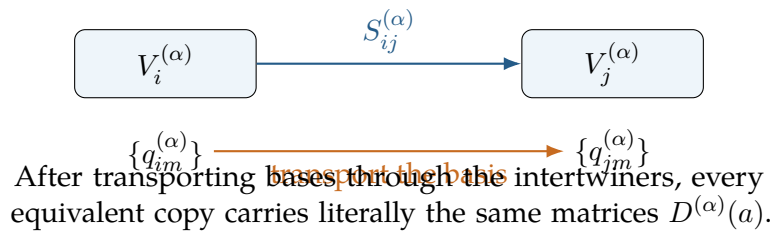


图 23.2: Equalising the matrix representatives on equivalent irreducible summands.

In the resulting basis $e_{im}^{(\alpha)}$, the matrix entries of the representation are

$$D_{ij;mn}^{(\alpha\beta)}(a) = \delta_{\alpha\beta} \delta_{ij} D_{mn}^{(\alpha)}(a).$$

The first pair (α, i) selects an irreducible block, while m selects a component inside that block.

23.3.2 The block-by-block commutator calculation

Write the most general Hamiltonian block from $V_j^{(\beta)}$ to $V_i^{(\alpha)}$ as $H_{ij}^{(\alpha\beta)}$. The equation $HD = DH$ gives

$$\sum_{\eta,k} H_{ik}^{(\alpha\eta)} D_{kj}^{(\eta\beta)}(a) = \sum_{\eta,k} D_{ik}^{(\alpha\eta)}(a) H_{kj}^{(\eta\beta)}.$$

Substituting the Kronecker deltas from (23.3.1) collapses both sums:

$$H_{ij}^{(\alpha\beta)} D^{(\beta)}(a) = D^{(\alpha)}(a) H_{ij}^{(\alpha\beta)} \quad (a \in G).$$

Thus every Hamiltonian block is an intertwiner.

If $\alpha \neq \beta$, the irreducible representations are inequivalent, so Schur's lemma gives

$$H_{ij}^{(\alpha\beta)} = 0.$$

If $\alpha = \beta$, the two matrices on the sides of (23.3.2) are the same irreducible representation. The complex scalar-commutant form of Schur's lemma gives

$$H_{ij}^{(\alpha\alpha)} = C_{ij}^{(\alpha)} I_{d_\alpha}$$

for a scalar $C_{ij}^{(\alpha)}$.

Theorem 23.2: Hamiltonians in a symmetry-adapted basis

Suppose

$$[H, D(a)] = 0 \quad (a \in G), \quad D \cong \bigoplus_{\alpha} n_{\alpha} D^{(\alpha)},$$

where the $D^{(\alpha)}$ are pairwise inequivalent finite-dimensional irreducible unitary complex representations. There is a basis, determined by D and independent of H , in which

$$H_{ij;mn}^{(\alpha\beta)} = \delta_{\alpha\beta} C_{ij}^{(\alpha)} \delta_{mn}.$$

Equivalently,

$$V \cong \bigoplus_{\alpha} (\mathbb{C}^{n_{\alpha}} \otimes W_{\alpha}), \quad D(a) = \bigoplus_{\alpha} (I_{n_{\alpha}} \otimes D^{(\alpha)}(a)),$$

and

$$H = \bigoplus_{\alpha} (C^{(\alpha)} \otimes I_{d_{\alpha}}).$$

When $H = H^{\dagger}$, every multiplicity-space matrix $C^{(\alpha)}$ is Hermitian. Indeed,

$$H^{\dagger} = \bigoplus_{\alpha} (C^{(\alpha)\dagger} \otimes I_{d_{\alpha}}),$$

so equality with H implies $C^{(\alpha)\dagger} = C^{(\alpha)}$. Diagonalise it by a unitary matrix U_{α} :

$$U_{\alpha}^{\dagger} C^{(\alpha)} U_{\alpha} = \text{diag}(E_{\alpha 1}, \dots, E_{\alpha n_{\alpha}}).$$

The change of basis $U_{\alpha} \otimes I_{d_{\alpha}}$ mixes only equivalent copies and therefore leaves $D(a)$ in the same

repeated-block form. Each eigenvalue $E_{\alpha r}$ occurs at least d_α times:

$$\underbrace{E_{\alpha r}, \dots, E_{\alpha r}}_{d_\alpha \text{ copies}}$$

This is the *symmetry-induced degeneracy*.

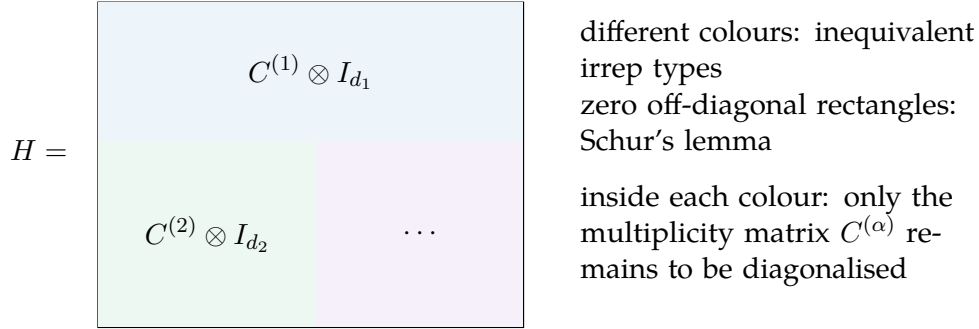


图 23.3: The universal block form of a symmetry-invariant Hamiltonian.

23.3.3 What symmetry does and does not determine

The total dimension is

$$\dim V = \sum_{\alpha} n_{\alpha} d_{\alpha}.$$

Without symmetry one diagonalises one matrix of that size. With symmetry, one diagonalises the smaller matrices $C^{(\alpha)}$, whose sizes are $n_{\alpha} \times n_{\alpha}$.

Representation data	Consequence for H	Physical meaning
$d_{\alpha} = \dim D^{(\alpha)}$	every eigenvalue from $C^{(\alpha)}$ is repeated at least d_{α} times	symmetry-enforced degeneracy
n_{α} , the number of equivalent copies	size of the matrix $C^{(\alpha)}$	remaining diagonalisation problem
all $d_{\alpha} = 1$	no degeneracy is forced	typical for finite-dimensional complex irreps of Abelian groups
all $n_{\alpha} = 1$	every $C^{(\alpha)}$ is a scalar	the symmetry basis already diagonalises H

Two eigenvalues may coincide even when symmetry does not require them to. For example, if $H = H(\lambda)$, levels from distinct irreducible sectors can cross as λ varies:

$$E_{\alpha r}(\lambda_0) = E_{\beta s}(\lambda_0).$$

Such a coincidence is called an *accidental degeneracy relative to G* .

Sometimes a larger group $\overline{G} \supset G$ explains the coincidence. The apparently distinct G -irreducible spaces may combine into a single larger-dimensional irreducible representation of $\overline{G}$. Relative to the larger group there are fewer sectors, larger d_{α} , and often smaller multiplicities n_{α} .

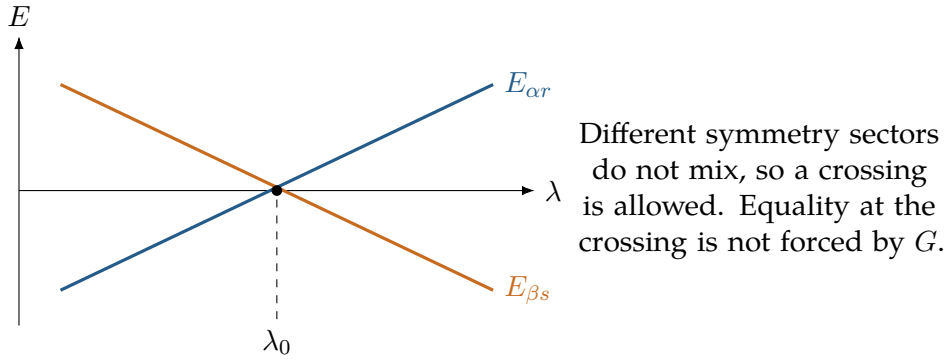


图 23.4: An accidental degeneracy at a parameter-dependent level crossing.

The hydrogenic equality across different ℓ values is the standard example. Thus “accidental” means unexplained by the symmetry currently being used, not necessarily devoid of a deeper cause.

A basis-independent statement

The decomposition

$$V \cong \bigoplus_{\alpha} \mathbb{C}^{n_{\alpha}} \otimes W_{\alpha}$$

is called the decomposition into *isotypic components*. The commutant of $D(G)$ is

$$\text{End}_G(V) \cong \bigoplus_{\alpha} M_{n_{\alpha}}(\mathbb{C}) \otimes I_{W_{\alpha}}.$$

The Hamiltonian theorem is precisely the statement that every G -invariant Hamiltonian belongs to this commutant.

23.4 The spin- $\frac{1}{2}$ representation

Let $V = \mathbb{C}^2$ with orthonormal basis $\{|+\rangle, |-\rangle\}$, where

$$|+\rangle = |\tfrac{1}{2}\rangle, \quad |-\rangle = |-\tfrac{1}{2}\rangle.$$

In units with $\hbar = 1$, define

$$S_{\alpha} = \frac{1}{2} \sum_{m, m' = \pm 1/2} |m\rangle (\sigma_{\alpha})_{mm'} \langle m'|, \quad \alpha = x, y, z,$$

where

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Direct multiplication gives

$$\sigma_{\alpha} \sigma_{\beta} = \delta_{\alpha\beta} I + i \sum_{\gamma} \varepsilon_{\alpha\beta\gamma} \sigma_{\gamma}.$$

Indeed, each Pauli matrix squares to I , while, for example,

$$\sigma_x \sigma_y = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} = i\sigma_z, \quad \sigma_y \sigma_x = -i\sigma_z.$$

Taking the difference of the two orders yields

$$[S_\alpha, S_\beta] = i \sum_{\gamma} \varepsilon_{\alpha\beta\gamma} S_\gamma.$$

The Casimir operator is

$$\begin{aligned} \mathbf{S}^2 &= S_x^2 + S_y^2 + S_z^2 \\ &= \frac{1}{4} (\sigma_x^2 + \sigma_y^2 + \sigma_z^2) \\ &= \frac{3}{4} I = s(s+1)I, \end{aligned}$$

so $s = \frac{1}{2}$. Also

$$S_z |\pm\rangle = \pm \frac{1}{2} |\pm\rangle.$$

For $\mathbf{a} = \phi \hat{\mathbf{n}} \in \mathbb{R}^3$, define

$$D^{(1/2)}(\mathbf{a}) = \exp(-i\mathbf{a} \cdot \mathbf{S}) = \exp\left(-\frac{i}{2} \mathbf{a} \cdot \boldsymbol{\sigma}\right).$$

Because

$$(\hat{\mathbf{n}} \cdot \boldsymbol{\sigma})^2 = I,$$

separating the exponential series into its even and odd powers gives the explicit formula

$$D^{(1/2)}(\hat{\mathbf{n}}, \phi) = \cos \frac{\phi}{2} I - i \sin \frac{\phi}{2} \hat{\mathbf{n}} \cdot \boldsymbol{\sigma}.$$

It is unitary because its exponent is anti-Hermitian. It also has determinant one:

$$\det D^{(1/2)}(\mathbf{a}) = \det \exp\left(-\frac{i}{2} \mathbf{a} \cdot \boldsymbol{\sigma}\right) = \exp\left(\operatorname{tr}\left[-\frac{i}{2} \mathbf{a} \cdot \boldsymbol{\sigma}\right]\right) = 1,$$

since every Pauli matrix is traceless. Thus these are matrices in $SU(2)$.

Proposition 23.1: Irreducibility of spin one-half

The defining two-dimensional representation $D^{(1/2)}$ of $SU(2)$ is irreducible.

证明. A proper nonzero invariant subspace of $\mathbb{C}^2$ would be one-dimensional, say $W = \mathbb{C}v$. Invariance under every $\exp(-itS_\alpha)$ implies, after differentiating at $t = 0$, that $S_\alpha v \in W$ for all α . Thus v would be a common eigenvector of σ_x and σ_z .

The eigenvectors of σ_z are multiples of $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$, but

$$\sigma_x \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad \sigma_x \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Neither σ_z -eigenline is invariant under σ_x . Hence no such W exists. ■

The spin operators transform as a three-dimensional vector:

$$D^{(1/2)}(\mathbf{a})S_\beta D^{(1/2)}(\mathbf{a})^\dagger = \sum_{\alpha=x,y,z} R_{\alpha\beta}(\mathbf{a})S_\alpha,$$

where $R(\mathbf{a}) \in SO(3)$ is the ordinary axis-angle rotation. To see the infinitesimal mechanism, put $X = -i\phi \hat{\mathbf{n}} \cdot \mathbf{S}$. Hadamard's lemma gives

$$e^X S_\beta e^{-X} = S_\beta + [X, S_\beta] + \frac{1}{2!}[X, [X, S_\beta]] + \cdots.$$

Using the spin commutation relations,

$$[X, S_\beta] = \phi \sum_{\alpha,\gamma} \hat{n}_\gamma \varepsilon_{\gamma\beta\alpha} S_\alpha.$$

This is exactly the infinitesimal generator of a real rotation acting on a three-vector. Exponentiating it gives (23.4).

23.5 Two spin- $\frac{1}{2}$ particles

Let

$$V = V_1 \otimes V_2, \quad \dim V = 4,$$

and let $\mathbf{S}_1 = \mathbf{S} \otimes I$ and $\mathbf{S}_2 = I \otimes \mathbf{S}$. Consider the isotropic exchange Hamiltonian

$$H = J \mathbf{S}_1 \cdot \mathbf{S}_2 = J \sum_{\alpha=x,y,z} S_{1\alpha} S_{2\alpha}.$$

The rotation representation on the product space is

$$D(\mathbf{a}) = D^{(1/2)}(\mathbf{a}) \otimes D^{(1/2)}(\mathbf{a}).$$

Because operators acting on distinct tensor factors commute,

$$\begin{aligned} D(\mathbf{a}) &= e^{-i\mathbf{a} \cdot \mathbf{S}_1} e^{-i\mathbf{a} \cdot \mathbf{S}_2} \\ &= e^{-i\mathbf{a} \cdot (\mathbf{S}_1 + \mathbf{S}_2)} = e^{-i\mathbf{a} \cdot \mathbf{S}_{\text{tot}}}, \end{aligned}$$

where $\mathbf{S}_{\text{tot}} = \mathbf{S}_1 + \mathbf{S}_2$.

The Hamiltonian is invariant. Using (23.4) on each factor,

$$\begin{aligned} D(\mathbf{a}) H D(\mathbf{a})^\dagger &= J \sum_{\alpha} \left(\sum_{\beta} R_{\beta\alpha} S_{1\beta} \right) \left(\sum_{\gamma} R_{\gamma\alpha} S_{2\gamma} \right) \\ &= J \sum_{\beta,\gamma} \left(\sum_{\alpha} R_{\beta\alpha} R_{\gamma\alpha} \right) S_{1\beta} S_{2\gamma} \\ &= J \sum_{\beta,\gamma} \delta_{\beta\gamma} S_{1\beta} S_{2\gamma} = H, \end{aligned}$$

where $RR^\top = I$ was used in the third line.

23.5.1 Product and coupled bases

The uncoupled orthonormal basis is

$$|m_1 m_2\rangle = |m_1\rangle \otimes |m_2\rangle, \quad m_1, m_2 = \pm \frac{1}{2}.$$

The coupled basis consists of simultaneous eigenvectors of S_{tot}^2 and $S_{\text{tot},z}$:

$$|S, M\rangle, \quad S = 0, 1, \quad M = -S, -S + 1, \dots, S.$$

The change of basis is expressed by Clebsch–Gordan coefficients:

$$|S, M\rangle = \sum_{m_1, m_2} |m_1 m_2\rangle \langle m_1 m_2 | S, M\rangle,$$

$$|m_1 m_2\rangle = \sum_{S, M} |S, M\rangle \langle S, M | m_1 m_2\rangle.$$

The coefficient vanishes unless $M = m_1 + m_2$.

Starting with the unique highest-weight vector and lowering it gives the coefficients explicitly:

$$\begin{aligned} |1, 1\rangle &= |++\rangle, \\ |1, 0\rangle &= \frac{1}{\sqrt{2}}(|+-\rangle + |-+\rangle), \\ |1, -1\rangle &= |--\rangle, \\ |0, 0\rangle &= \frac{1}{\sqrt{2}}(|+-\rangle - |-+\rangle). \end{aligned}$$

The first three symmetric states form the triplet V_1 ; the last antisymmetric state forms the singlet V_0 . Therefore

$$D^{(1/2)} \otimes D^{(1/2)} = D^{(0)} \oplus D^{(1)}, \quad \frac{1}{2} \otimes \frac{1}{2} = 0 \oplus 1.$$

There are no matrix elements between the $S = 0$ and $S = 1$ spaces, because they are inequivalent irreducible representations.

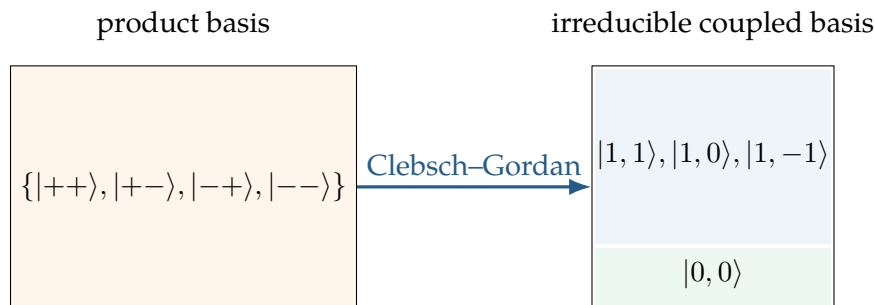


图 23.5: The product representation becomes a spin-one block and a spin-zero block in the coupled basis.

23.5.2 The spectrum without a 4×4 determinant

Since

$$S_{\text{tot}}^2 = S_1^2 + S_2^2 + 2S_1 \cdot S_2,$$

we have

$$\mathbf{S}_1 \cdot \mathbf{S}_2 = \frac{1}{2} (\mathbf{S}_{\text{tot}}^2 - \mathbf{S}_1^2 - \mathbf{S}_2^2).$$

On the subspace of total spin S ,

$$\begin{aligned} E_S &= \frac{J}{2} \left[S(S+1) - \frac{1}{2} \left(\frac{1}{2} + 1 \right) - \frac{1}{2} \left(\frac{1}{2} + 1 \right) \right] \\ &= \frac{J}{2} \left[S(S+1) - \frac{3}{2} \right]. \end{aligned}$$

Consequently,

$$E_0 = -\frac{3J}{4} \quad (\text{one singlet state}), \quad E_1 = \frac{J}{4} \quad (\text{three triplet states}).$$

For $J > 0$ the singlet lies below the triplet. For $J < 0$ the order is reversed: the threefold-degenerate triplet lies below the singlet.

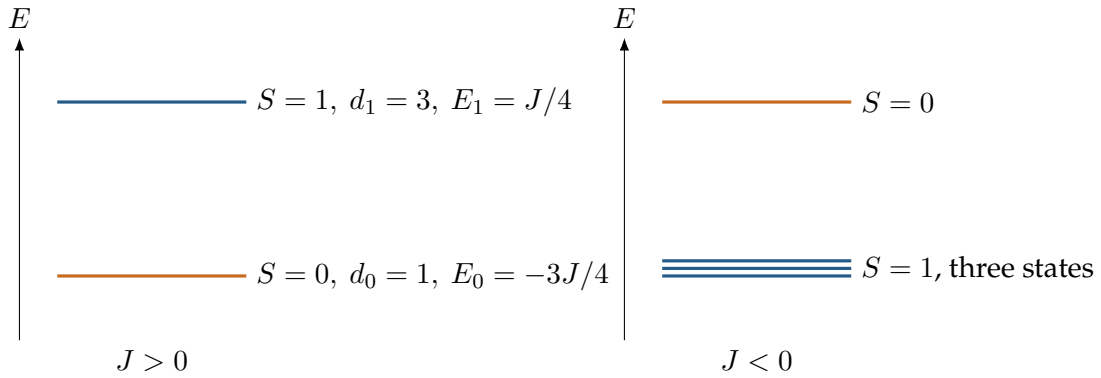


图 23.6: Singlet and triplet levels for the two signs of the exchange coupling. The closely drawn lines at right represent three states at the same energy, not three distinct energies.

23.5.3 General spin- j matrices

For any $j = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots$, the Wigner matrices are

$$D_{m'm}^{(j)}(\mathbf{a}) = \langle j, m' | \exp(-i\mathbf{a} \cdot \mathbf{J}) | j, m \rangle, \quad m, m' = -j, \dots, j.$$

The Casimir $\mathbf{J}^2$ commutes with all J_α , so rotations cannot mix different j -subspaces. Within a fixed j -subspace, the raising and lowering operators

$$J_\pm = J_x \pm iJ_y$$

connect every weight state:

$$J_\pm |j, m\rangle = \sqrt{(j \mp m)(j \pm m + 1)} |j, m \pm 1\rangle.$$

Any nonzero invariant subspace containing one weight vector therefore contains all $2j + 1$ weight vectors. Hence $D^{(j)}$ is irreducible.

23.6 Three and more spin- $\frac{1}{2}$ particles

The general angular-momentum addition rule is

$$s_1 \otimes s_2 = \bigoplus_{S=|s_1-s_2|}^{s_1+s_2} S,$$

where S increases in integer steps. This is the representation-theoretic form of the triangle inequality

$$|s_1 - s_2| \leq S \leq s_1 + s_2.$$

23.6.1 Three spins and a nontrivial multiplicity matrix

For three spin- $\frac{1}{2}$ spaces,

$$\begin{aligned} \left(\frac{1}{2}\right)^{\otimes 3} &= (0 \oplus 1) \otimes \frac{1}{2} \\ &= \frac{1}{2} \oplus \left(\frac{1}{2} \oplus \frac{3}{2}\right) \\ &= 2 \left(\frac{1}{2}\right) \oplus \frac{3}{2}. \end{aligned}$$

Thus

$$n_{1/2} = 2, \quad d_{1/2} = 2, \quad n_{3/2} = 1, \quad d_{3/2} = 4.$$

Every rotationally invariant Hamiltonian has the structure

$$H = \left(C^{(1/2)} \otimes I_2\right) \oplus \left(E_{3/2} I_4\right),$$

where $C^{(1/2)}$ is a 2×2 Hermitian matrix. The quartet energy is fourfold degenerate, and each eigenvalue of $C^{(1/2)}$ is a doublet.

Consider the open three-site exchange Hamiltonian from the handwritten notes,

$$H = J(\mathbf{S}_1 \cdot \mathbf{S}_2 + \mathbf{S}_2 \cdot \mathbf{S}_3).$$

It commutes with the total-spin representation because each dot product is a rotational scalar. The abstract block form already determines all degeneracies. To exhibit the remaining 2×2 problem explicitly, couple spins 1 and 2 first. In the $M = +\frac{1}{2}$ sector choose

$$\begin{aligned} |a\rangle &= |s_{12} = 0; S = \frac{1}{2}, M = \frac{1}{2}\rangle = \frac{1}{\sqrt{2}} (|+-+\rangle - |-++\rangle), \\ |b\rangle &= |s_{12} = 1; S = \frac{1}{2}, M = \frac{1}{2}\rangle \\ &= \sqrt{\frac{2}{3}} |++-\rangle - \frac{1}{\sqrt{6}} (|+-+\rangle + |-++\rangle). \end{aligned}$$

For any pair of spin- $\frac{1}{2}$ particles,

$$\begin{aligned} \mathbf{S}_i \cdot \mathbf{S}_j |++\rangle &= \frac{1}{4} |++\rangle, \\ \mathbf{S}_i \cdot \mathbf{S}_j |+-\rangle &= -\frac{1}{4} |+-\rangle + \frac{1}{2} |-+\rangle, \\ \mathbf{S}_i \cdot \mathbf{S}_j |-+\rangle &= -\frac{1}{4} |-+\rangle + \frac{1}{2} |+-\rangle, \\ \mathbf{S}_i \cdot \mathbf{S}_j |--\rangle &= \frac{1}{4} |--\rangle. \end{aligned}$$

Applying these four elementary rules to the states above gives

$$C^{(1/2)} = J \begin{pmatrix} -\frac{3}{4} & \frac{\sqrt{3}}{4} \\ \frac{\sqrt{3}}{4} & -\frac{1}{4} \end{pmatrix}.$$

Its characteristic polynomial is

$$\begin{aligned} \det(C^{(1/2)} - EI_2) &= \left(-\frac{3J}{4} - E\right) \left(-\frac{J}{4} - E\right) - \frac{3J^2}{16} \\ &= E(E + J). \end{aligned}$$

Hence the two doublet energies are

$$E_{1/2}^{(1)} = -J, \quad E_{1/2}^{(2)} = 0.$$

On the totally symmetric $S = \frac{3}{2}$ space, each neighbouring pair is a triplet and contributes $J/4$, so

$$E_{3/2} = \frac{J}{2}.$$

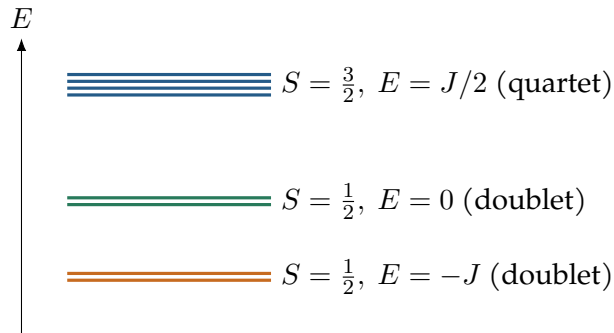


图 23.7: Spectrum of the open three-spin Hamiltonian for $J > 0$. The multiplicities 2, 2, 4 are dictated by rotation symmetry.

23.6.2 Four spins and the coupling tree

Adding one more spin gives

$$\begin{aligned} \left(\frac{1}{2}\right)^{\otimes 4} &= \left[2\left(\frac{1}{2}\right) \oplus \frac{3}{2}\right] \otimes \frac{1}{2} \\ &= 2(0 \oplus 1) \oplus (1 \oplus 2) \\ &= 2(0) \oplus 3(1) \oplus 1(2). \end{aligned}$$

The dimension check is

$$2 \cdot 1 + 3 \cdot 3 + 1 \cdot 5 = 2 + 9 + 5 = 16 = 2^4.$$

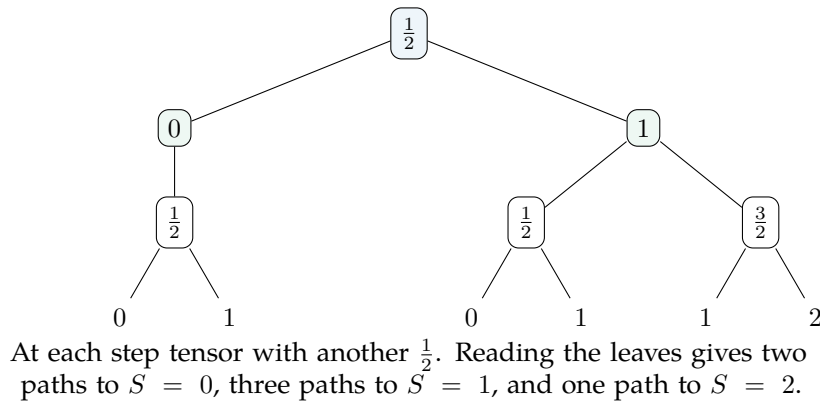


图 23.8: A coupling tree for four spin- $\frac{1}{2}$ particles. Different paths ending at the same S are the equivalent copies counted by n_S .

23.6.3 The half-Pascal triangle and its closed formula

Let $n_S(L)$ be the multiplicity of total spin S in $(\frac{1}{2})^{\otimes L}$. Adding the last spin and using the triangle rule shows that a final spin S can arise from $S - \frac{1}{2}$ or $S + \frac{1}{2}$ at the previous stage:

$$n_S(L+1) = n_{S-1/2}(L) + n_{S+1/2}(L),$$

with nonexistent or negative-spin multiplicities interpreted as zero. The first seven rows are

$L \backslash S$	0	$\frac{1}{2}$	1	$\frac{3}{2}$	2	$\frac{5}{2}$	3	$\frac{7}{2}$
1		1						
2	1		1					
3		2		1				
4	2		3		1			
5		5		4		1		
6	5		9		5		1	
7		14		14		6		1

This is the “half-Pascal triangle” drawn in the notes.

There is also a closed formula:

$$n_S(L) = \binom{L}{L/2 - S} - \binom{L}{L/2 - S - 1}.$$

Here a binomial coefficient is zero if its lower argument is not an integer between 0 and L .

证明. Fix a total magnetic quantum number M . A product state has

$$M = \frac{1}{2}(N_+ - N_-) = \frac{L}{2} - N_-.$$

Therefore the number of product-basis states with that M is

$$N_M = \binom{L}{L/2 - M}.$$

Every copy of a spin- S' irreducible representation contains exactly one state of magnetic quantum number M whenever $S' \geq |M|$. For $M \geq 0$,

$$N_M = \sum_{S' \geq M} n_{S'}(L).$$

Subtract the equation at $M = S + 1$ from the equation at $M = S$:

$$\begin{aligned} n_S(L) &= N_S - N_{S+1} \\ &= \binom{L}{L/2 - S} - \binom{L}{L/2 - S - 1}. \end{aligned}$$

This proves the formula. ■

As a final consistency check, the irreducible dimensions must recover the full tensor-product dimension:

$$\sum_S n_S(L)(2S + 1) = 2^L.$$

The left side counts each spin- S copy with its $2S + 1$ magnetic components, while the right side counts all choices of $+$ and $-$ on the L sites.

23.7 Tight-binding model on an open chain

Let

$$V = \text{span}\{e_1, \dots, e_L\}, \quad \langle e_i, e_j \rangle = \delta_{ij},$$

where $e_i = |i\rangle$ is an orbital localised at site i . A general tight-binding Hamiltonian is

$$H = \sum_{i,j=1}^L T_{ij} |i\rangle \langle j| = \sum_{i,j=1}^L T_{ij} e_i e_j^\dagger.$$

Hermiticity requires $T_{ji} = T_{ij}^*$. For a nearest-neighbour open chain with real uniform hopping T ,

$$H_{\text{open}} = T \sum_{i=1}^{L-1} (|i\rangle\langle i+1| + |i+1\rangle\langle i|).$$

Its matrix is tridiagonal:

$$H_{\text{open}} = T \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 1 & 0 & 1 & \ddots & \vdots \\ 0 & 1 & 0 & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & 1 \\ 0 & \cdots & 0 & 1 & 0 \end{pmatrix}.$$

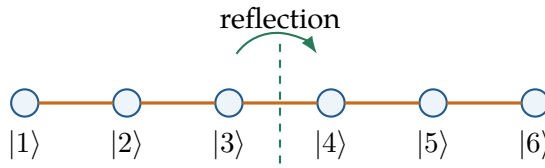


图 23.9: The open six-site chain and its only nontrivial spatial symmetry.

Let s reflect the chain:

$$D_{\text{chain}}(s)e_i = e_{L+1-i}, \quad D_{\text{chain}}(e) = I.$$

This is a unitary representation of $\mathbb{Z}_2$. In the site basis its nontrivial matrix is the anti-diagonal permutation matrix

$$D_{\text{chain}}(s) = \begin{pmatrix} 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & \cdots & 1 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 1 & \cdots & 0 & 0 \\ 1 & 0 & \cdots & 0 & 0 \end{pmatrix}.$$

Reflection maps every nearest-neighbour bond $(i, i+1)$ to the bond $(L-i, L+1-i)$, so

$$D_{\text{chain}}(s)H_{\text{open}}D_{\text{chain}}(s)^{-1} = H_{\text{open}}.$$

For even L , define

$$f_j^+ = \frac{e_j + e_{L+1-j}}{\sqrt{2}}, \quad f_j^- = \frac{e_j - e_{L+1-j}}{\sqrt{2}}, \quad j = 1, \dots, \frac{L}{2}.$$

Then

$$D_{\text{chain}}(s)f_j^+ = f_j^+, \quad D_{\text{chain}}(s)f_j^- = -f_j^-.$$

For every j , the line $\mathbb{C}f_j^+$ carries the even irrep. The line $\mathbb{C}f_j^-$ carries the odd irrep. Thus

$$D_{\text{chain}} = \frac{L}{2} D^{(+)} \oplus \frac{L}{2} D^{(-)} \quad (L \text{ even}).$$

For odd L , the fixed middle site is even, so the multiplicities are $(L+1)/2$ and $(L-1)/2$, respectively.

For the $L = 6$ example drawn in the notes, the invariant lines are

even	odd
$\text{span}\{e_1 + e_6\}$	$\text{span}\{e_1 - e_6\}$
$\text{span}\{e_2 + e_5\}$	$\text{span}\{e_2 - e_5\}$
$\text{span}\{e_3 + e_4\}$	$\text{span}\{e_3 - e_4\}$

and the symmetry-adapted Hamiltonian is

$$H_{\text{open}} \cong H_+ \oplus H_-,$$

with

$$H_+ = T \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}, \quad H_- = T \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & -1 \end{pmatrix}.$$

For example, the $+1$ in the last diagonal entry of H_+ and the -1 in that of H_- come from the central bond $|3\rangle\langle 4| + |4\rangle\langle 3|$.

All complex irreducible representations of $\mathbb{Z}_2$ are one-dimensional. Hence $d_+ = d_- = 1$: parity creates two smaller diagonalisation problems but forces no degeneracy. This can be checked by solving the chain completely. Put

$$\psi_n = \sqrt{\frac{2}{L+1}} \sum_{j=1}^L \sin\left(\frac{n\pi j}{L+1}\right) e_j, \quad n = 1, \dots, L.$$

Using

$$\sin((j-1)q) + \sin((j+1)q) = 2 \cos q \sin(jq)$$

and the boundary values $\sin 0 = \sin(n\pi) = 0$, one obtains

$$H_{\text{open}} \psi_n = 2T \cos\left(\frac{n\pi}{L+1}\right) \psi_n.$$

The cosine is strictly decreasing on $(0, \pi)$, so these L energies are distinct. Reflection gives

$$\begin{aligned} \sin\left(\frac{n\pi(L+1-j)}{L+1}\right) &= \sin\left(n\pi - \frac{n\pi j}{L+1}\right) \\ &= (-1)^{n+1} \sin\left(\frac{n\pi j}{L+1}\right), \end{aligned}$$

and therefore

$$D_{\text{chain}}(s) \psi_n = (-1)^{n+1} \psi_n.$$

The exact eigenvectors alternate between the two parity sectors.

23.8 Tight-binding model on a ring

Close the chain by identifying $|L+1\rangle = |1\rangle$:

$$H_{\text{ring}} = T \sum_{i=1}^L (|i\rangle\langle i+1| + |i+1\rangle\langle i|).$$

The spatial symmetry is now the dihedral group

$$D_L = \langle r, s \mid r^L = e, s^2 = e, srs = r^{-1} \rangle,$$

of order $2L$. On the site basis, choose the generators

$$D_{\text{ring}}(r)e_i = e_{i-1}, \quad D_{\text{ring}}(s)e_i = e_{2-i},$$

where indices are read modulo L , with $e_0 = e_L$. The first generator is one step around the ring; the second is reflection about the axis through site 1. Both merely permute the bonds in (23.8), so

$$[H_{\text{ring}}, D_{\text{ring}}(a)] = 0 \quad (a \in D_L).$$

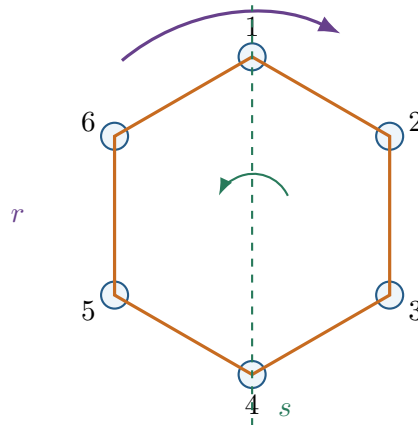


图 23.10: The six-site ring. Rotations and reflections generate the non-Abelian group D_6 .

23.8.1 Real Fourier subspaces

Put

$$\theta_k = \frac{2\pi k}{L}.$$

The constant vector

$$f_0 = \frac{1}{\sqrt{L}} \sum_{i=1}^L e_i$$

spans the trivial representation. When L is even, the alternating vector

$$f_{L/2} = \frac{1}{\sqrt{L}} \sum_{i=1}^L (-1)^i e_i$$

spans another one-dimensional representation:

$$D(r)f_{L/2} = -f_{L/2}, \quad D(s)f_{L/2} = f_{L/2}.$$

For

$$k = 1, \dots, \left\lfloor \frac{L-1}{2} \right\rfloor,$$

define

$$c_k = \sqrt{\frac{2}{L}} \sum_{i=1}^L \cos(\theta_k i) e_i,$$

$$s_k = \sqrt{\frac{2}{L}} \sum_{i=1}^L \sin(\theta_k i) e_i.$$

The elementary discrete orthogonality identities for sines and cosines show that all these vectors form an orthonormal basis.

We now compute the group action rather than merely quote it. Because $D(r)e_i = e_{i-1}$, relabelling $j = i - 1$ gives

$$\begin{aligned} D(r)c_k &= \sqrt{\frac{2}{L}} \sum_i \cos(\theta_k i) e_{i-1} \\ &= \sqrt{\frac{2}{L}} \sum_j \cos(\theta_k (j+1)) e_j \\ &= \cos \theta_k c_k - \sin \theta_k s_k. \end{aligned}$$

The same calculation with sine gives

$$D(r)s_k = \sin \theta_k c_k + \cos \theta_k s_k.$$

Thus, with column-vector coordinates in the ordered basis (c_k, s_k) ,

$$D^{(k)}(r) = \begin{pmatrix} \cos \theta_k & \sin \theta_k \\ -\sin \theta_k & \cos \theta_k \end{pmatrix}.$$

If the opposite convention $D(r)e_i = e_{i+1}$ is chosen, this matrix is transposed; the representation is equivalent and all conclusions are unchanged.

For the reflection, relabel $j = 2 - i$:

$$\begin{aligned} D(s)c_k &= \sqrt{\frac{2}{L}} \sum_j \cos(\theta_k (2-j)) e_j \\ &= \cos(2\theta_k) c_k + \sin(2\theta_k) s_k, \\ D(s)s_k &= \sin(2\theta_k) c_k - \cos(2\theta_k) s_k. \end{aligned}$$

Therefore

$$D^{(k)}(s) = \begin{pmatrix} \cos(2\theta_k) & \sin(2\theta_k) \\ \sin(2\theta_k) & -\cos(2\theta_k) \end{pmatrix}.$$

This matrix squares to I_2 , while direct multiplication verifies

$$D^{(k)}(s)D^{(k)}(r)D^{(k)}(s) = D^{(k)}(r)^{-1}.$$

23.8.2 The complete $L = 6$ calculation

For the six-site example in the handwritten pages, use the ordered subspaces

subspace	orthonormal basis	dimension
V_1	$f_1 = \frac{1}{\sqrt{6}} \sum_{i=1}^6 e_i$	1
V_2	$f_2 = \frac{1}{\sqrt{6}} \sum_{i=1}^6 (-1)^i e_i$	1
V_3	$f_3 = \frac{1}{\sqrt{3}} \sum_i \cos\left(\frac{2\pi i}{3}\right) e_i, \quad f_4 = \frac{1}{\sqrt{3}} \sum_i \sin\left(\frac{2\pi i}{3}\right) e_i$	2
V_4	$f_5 = \frac{1}{\sqrt{3}} \sum_i \cos\left(\frac{\pi i}{3}\right) e_i, \quad f_6 = \frac{1}{\sqrt{3}} \sum_i \sin\left(\frac{\pi i}{3}\right) e_i$	2

The one-dimensional actions are

$$\begin{array}{c|cc} & r & s \\ \hline D^{(1)} & 1 & 1 \\ D^{(2)} & -1 & 1 \end{array}.$$

For V_3 , where $\theta = 2\pi/3$,

$$D^{(3)}(r) = \begin{pmatrix} \cos \frac{2\pi}{3} & \sin \frac{2\pi}{3} \\ -\sin \frac{2\pi}{3} & \cos \frac{2\pi}{3} \end{pmatrix},$$

$$D^{(3)}(s) = \begin{pmatrix} \cos \frac{2\pi}{3} & -\sin \frac{2\pi}{3} \\ -\sin \frac{2\pi}{3} & -\cos \frac{2\pi}{3} \end{pmatrix}.$$

This second matrix follows from the general reflection formula, with $2\theta = 4\pi/3$. In writing it we used $\cos(4\pi/3) = \cos(2\pi/3)$ and $\sin(4\pi/3) = -\sin(2\pi/3)$.

For V_4 , where $\theta = \pi/3$,

$$D^{(4)}(r) = \begin{pmatrix} \cos \frac{\pi}{3} & \sin \frac{\pi}{3} \\ -\sin \frac{\pi}{3} & \cos \frac{\pi}{3} \end{pmatrix},$$

$$D^{(4)}(s) = \begin{pmatrix} -\cos \frac{\pi}{3} & \sin \frac{\pi}{3} \\ \sin \frac{\pi}{3} & \cos \frac{\pi}{3} \end{pmatrix}.$$

The two-dimensional representations are not equivalent. Equivalent matrices have equal traces, but

$$\text{tr } D^{(3)}(r) = 2 \cos \frac{2\pi}{3} = -1, \quad \text{tr } D^{(4)}(r) = 2 \cos \frac{\pi}{3} = 1.$$

This is the trace test carried out in the notes. Consequently,

$$D_{\text{ring}} = D^{(1)} \oplus D^{(2)} \oplus D^{(3)} \oplus D^{(4)} \quad (L = 6),$$

and all four summands are mutually inequivalent.

For completeness, the handwritten final-page calculation of the V_3 rotation is reproduced in

one uninterrupted chain:

$$\begin{aligned}
 D^{(3)}(r)f_3 &= \frac{1}{\sqrt{3}} \sum_i \cos\left(\frac{2\pi i}{3}\right) D(r)e_i \\
 &= \frac{1}{\sqrt{3}} \sum_j \cos\left(\frac{2\pi(j+1)}{3}\right) e_j \\
 &= \frac{1}{\sqrt{3}} \sum_j \left[\cos\left(\frac{2\pi j}{3}\right) \cos \frac{2\pi}{3} - \sin\left(\frac{2\pi j}{3}\right) \sin \frac{2\pi}{3} \right] e_j \\
 &= \cos \frac{2\pi}{3} f_3 - \sin \frac{2\pi}{3} f_4.
 \end{aligned}$$

Analogously,

$$D^{(3)}(r)f_4 = \sin \frac{2\pi}{3} f_3 + \cos \frac{2\pi}{3} f_4,$$

which gives exactly the matrix stated above.

23.8.3 Irreducible decomposition and energy degeneracies

For even L , the vertex representation decomposes as

$$D_{\text{ring}} = D^{(0)} \oplus D^{(L/2)} \oplus \bigoplus_{k=1}^{L/2-1} D^{(k)},$$

The first two summands are one-dimensional. Every $D^{(k)}$ in the sum is two-dimensional. For odd L , only $D^{(0)}$ is one-dimensional and

$$D_{\text{ring}} = D^{(0)} \oplus \bigoplus_{k=1}^{(L-1)/2} D^{(k)}.$$

The two-dimensional spaces are irreducible: over $\mathbb{C}$, the rotation has eigenlines with eigenvalues $e^{\pm i\theta_k}$, but reflection interchanges those lines, so neither is invariant under the full dihedral group. The trace

$$\text{tr } D^{(k)}(r) = 2 \cos \theta_k$$

also shows that the listed summands are pairwise inequivalent.

Every summand appears once. Therefore the general symmetry theorem says that H_{ring} is already scalar on each listed subspace. The scalar can be found directly:

$$\begin{aligned}
 H_{\text{ring}}c_k &= T \sqrt{\frac{2}{L}} \sum_i \cos(\theta_k i) (e_{i-1} + e_{i+1}) \\
 &= 2T \cos \theta_k c_k,
 \end{aligned}$$

and similarly

$$H_{\text{ring}}s_k = 2T \cos \theta_k s_k.$$

Thus

$$E_k = 2T \cos \left(\frac{2\pi k}{L} \right).$$

The modes k and $L - k$, combined in the real basis (c_k, s_k) , have the same energy. Their twofold degeneracy is forced by the two-dimensional irreducible representation of D_L .

For $L = 6$, the complete result is

subspace	k	E_k	degeneracy
V_1	0	$2T$	1
V_4	1	T	2
V_3	2	$-T$	2
V_2	3	$-2T$	1

and the Hamiltonian in the symmetry basis has the schematic form

$$H_{\text{ring}} \cong (2T) \oplus (-2T) \oplus (-TI_2) \oplus (TI_2).$$

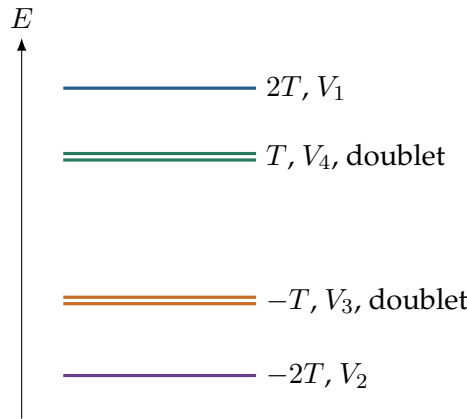


图 23.11: The six-site ring spectrum for $T > 0$. The two middle levels are doubly degenerate because V_3 and V_4 are two-dimensional irreducible D_6 -spaces.

The case $L = 4q + 2$, such as $L = 6$, has no zero-energy mode because no integer k satisfies $2\pi k/L = \pi/2$. If $L = 4q$, the allowed mode $k = L/4$ instead gives

$$E_{L/4} = 0$$

and belongs to a two-dimensional dihedral irrep, so the zero energy is twofold degenerate. This is the distinction between the $L = 4q + 2$ and $L = 4q$ rings noted in the lecture.

Why reflections matter

If one uses only the cyclic translation group C_L , all complex irreducible representations are one-dimensional. The two Fourier modes $e^{i\theta_k i}$ and $e^{-i\theta_k i}$ are then distinct one-dimensional sectors. Enlarging the symmetry to the full dihedral group adds a reflection that exchanges them; together they form one two-dimensional irreducible space, and the equality of their energies is then visibly symmetry-enforced.

What to retain from this chapter

1. Wigner's theorem represents transition-probability-preserving transformations by unitary or antiunitary operators, up to phase.
2. A unitary symmetry D of a Hamiltonian satisfies $D(a)HD(a)^{-1} = H$. The concrete physical action of D , not merely the abstract group, is essential.

3. If

$$D \cong \bigoplus_{\alpha} n_{\alpha} D^{(\alpha)},$$

then a symmetry-adapted basis gives

$$H = \bigoplus_{\alpha} \left(C^{(\alpha)} \otimes I_{d_{\alpha}} \right).$$

This follows by applying Schur's lemma to every Hamiltonian block.

4. The irrep dimension d_{α} is the minimum symmetry-induced degeneracy. The multiplicity n_{α} is the size of the matrix $C^{(\alpha)}$ still to be diagonalised.
5. A degeneracy unexplained by the chosen group is accidental relative to that group; a larger symmetry can turn it into an ordinary irrep degeneracy.
6. The spin- $\frac{1}{2}$ representation is generated by $S_{\alpha} = \sigma_{\alpha}/2$, and

$$\frac{1}{2} \otimes \frac{1}{2} = 0 \oplus 1.$$

The two-spin Heisenberg Hamiltonian therefore has a singlet and a triplet whose three states have equal energy.

7. For three spins,

$$\left(\frac{1}{2} \right)^{\otimes 3} = 2 \left(\frac{1}{2} \right) \oplus \frac{3}{2},$$

so an invariant Hamiltonian contains a 2×2 multiplicity matrix for the doublets and one scalar for the quartet.

8. The multiplicity of total spin S among L spin- $\frac{1}{2}$ particles is

$$n_S(L) = \binom{L}{L/2 - S} - \binom{L}{L/2 - S - 1}.$$

9. Reflection symmetry splits an open chain into even and odd blocks but forces no degeneracy, because the irreps of $\mathbb{Z}_2$ are one-dimensional.
10. The dihedral vertex representation of a ring contains irreducible Fourier subspaces of dimension two. On each such subspace,

$$E_k = 2T \cos \left(\frac{2\pi k}{L} \right)$$

is therefore twofold degenerate.